

RELATIONSHIP BETWEEN THE FUNCTIONAL STATUS OF PENILE CHOLINERGIC NERVES AND ERECTILE IMPOTENCE IN INSULIN DEPENDENT AND NON- INSULIN DEPENDENT DIABETIC MEN WITH AND WITHOUT NEUROPATHY

SYED TABREZ ALI

Laboratory of Molecular Neurobiology
Division of Experimental Therapeutics
Army Medical Centre, Washington, D.C. U.S.A.

ABSTRACT:

The functional status of penile cholinergic nerves in samples of corpus cavernosum was assessed by determining the accumulation of tritiated choline and the synthesis and release of tritiated-acetylcholine after incubation of corporeal tissue with tritiated-choline in 60 diabetic impotent men both insulin dependent (IDDM) and non-insulin dependent (NIDDM) with and without objective evidence of neuropathy and in 15 age-matched non diabetic impotent controls under-going surgery for the implantation of penile prosthesis. Both IDDM and NIDDM diabetic men with neuropathy exhibited significant decrease ($p < 0.05$) in the accumulation of tritiated choline and tritiated-acetylcholine synthesis compared to that from non-neuropathic diabetics (both IDDM and NIDDM) and non diabetic controls. However these values in both types of diabetic without neuropathy were not significantly different from those of non diabetic controls. These results provide evidence that an underlying defective neuropathic mechanism in the cholinergic nerves in corpus cavernosum of diabetic neuropathic men may play an etiological role in the pathogenesis of erectile impotence in these patients and may contribute to the development of penile-neuropathy irrespective of their type of diabetes.

Key words: Diabetes, neuropathy, cholinergic nerves, erectile impotence.

INTRODUCTION

Erectile impotence has been reported to be associated with diabetes mellitus (Ellenberg, 1980). The etiology has been reported to be a neuropathic abnormality in the male genital organs (Lehman and Jacobs, 1982) and/or vascular changes in the corpora cavernosa (Ali et al., 1993, Bart et al., 1994, Laung Khot et al., 1992, Weiss et al., 1972). Recently it has been recognized that it is likely that both sympathetic and parasympathetic nervous systems function together to produce highly specific vasodilatation and vasoconstriction responses in the appropriate vasculature (deGroat

Correspondence:

et al., 1980, Korenman and Viosca, 1993) that occur in erection. In addition, the contractile state of the smooth muscle of the trabeculae of the corpus cavernosum has been implicated to have a role in both erection and detumescence and this is also under autonomic nervous control (Willis et al., 1980, Diederichs et al., 1990, Knispal et al., 1992). Relaxation of smooth muscle of the arteries and trabeculae of the corpora cavernosa necessary for penile erection is under the control of non-adrenergic (Lue and Tanogho, 1987, Christ et al., 1992) non-cholinergic as well as cholinergic nerves (Saenz et al., 1982). Vasoactive intestinal polypeptide, the putative non-adrenergic, non cholinergic neurotransmitter, appears to act directly on the penile smooth muscle to cause relaxation (Hedlund and Andersson, 1985, Wang et al., 1993). Acetylcholine, the cholinergic neurotransmitter is probably an inhibitory modulator of adrenergic constrictor nerves and a facilitatory modulator of nor adrenergic nor cholinergic relaxation (Saenz et al., 1988, Azadzoi, 1992).

In the recent years many histochemical fluorescence techniques have demonstrated that vaso-active intestinal polypeptide (VIP) acts as transmitter in the nerves supplying the penile arteries (Anderson, 1984) and cavernous smooth muscles (Gu et al., 1983, Wang et al., 1993a). However to date there is no clinical electro-physiological technique for the evaluation of the integrity of these nerve fibers and the assessment of a neuropathic factor has been impeded by the lack of an objective laboratory test. Involvement of acetylcholine as a successful treatment for impaired neurogenic and endothelium mediated relaxation of penile smooth muscle from diabetic men with impotence has been reported by Saenz et al. (1988). An effect of intra-cavernous simultaneous injection of acetylcholine and vasoactive intestinal polypeptide on canine penile erection has also been reported (Yashiasta et al., 1992). These workers found that VIP increases affinity of muscarinic receptors for acetylcholine in canine corpus cavernosum. The levels of nor- adrenaline were also found to be higher in patients with psychogenic impotence than in normal controls and in patients with vasculogenic impotence (Sea and Moon, 1992).

In the present study we have used biochemical techniques to investigate the functional status of penile cholinergic nerves in IDDM and NIDDM diabetic impotent males with and without an objective evidence of neuropathy, which may be useful in defining diabetes-induced penile neuropathy in these patients.

MATERIALS AND METHODS

Specimens of human penile tissue containing corpus cavernosum were obtained from a series of 75 patients undergoing surgery for the implantation of penile prosthesis by a technique described previously (Seanz de Tajada et al., 1988). The series consisted of patients suffering from impotence due to a variety of causes

including diabetes, peyronie's disease and neuropathy. The cause of erectile dysfunction was categorized on the basis of clinical history, physical examination, sensory vibration perception threshold, psychiatric evaluation and additional evidence of neuropathy. For the experimental purposes a detailed questioner was completed which included the patient's age, weight and standard weight (Metropolitan life insurance company USA), age at the onset of diabetes and marital status. Diabetic treatment was recorded as diet alone, oral hypoglycemic agents or insulin. Inquiry was made of other drug therapy, angina pectoris, previous myocardial infarction or cardiac failure, intermittent claudication, thyroid function, previous sympathectomy or other abnormality that might predisposed to organic impotence such as neurological disease or previous injury. Careful inquiry was made of symptoms suggestive of peripheral neuropathy (numbness, paraesthesia, burning pain, sensory loss and weakness in lower limb, absence of ankle jerk, diminished sensation in the feet and/or muscle wasting and the evidence of recent or old neuropathic ulceration. The criteria for the presence of symptomatic peripheral neuropathy were one of more severe or two or more mild/moderate symptoms along with objective evidence (as outlined above) that these symptoms due to neuropathy. Inquiry was also made of symptoms of autonomic neuropathy which include postural hypotension (a fall of at least 30 mm Hg in systolic arterial blood pressure in upright position), intermittent diarrhea (characterized by episodes of fecal incontinence and nocturnal bouts), epigastric fullness (shown by contrast radiography) diminished sweating in legs, gustatory sweating and hypoglycemic unawareness (Table 1). The criteria for the presence of symptomatic autonomic neuropathy were two or more severe or three or more mild/moderate features. All such patients had well defined objective evidence of autonomic neuropathy from previous studies. The control subjects were hospitalized patients without glycosuria or post absorptive blood glucose levels above 5.5 m mol/L. None had neurological abnormality at physical examination. All the subjects were divided into 5 groups according to their clinical history and the evidence of vascular or neurological disorders. Four groups consisted of 15 patients each with IDDM and NIDDM diabetic impotence with and without objective evidence of neuropathy. The fifth group consisted of 15 non- diabetic impotent men who did not have any systemic or neurological disease affecting peripheral nerve. The mean age of patients in all the groups studied were comparable. Impotence was defined as constant inability to achieve or sustain an erection for up to 2 years.

Each tissue specimen was divided into several portions for analysis. Tissues were kept in chilled (4°C) Dulbecco's modified media and studied within 24 hours or were kept frozen and stored in liquid nitrogen. Corporeal tissue strips were incubated according to the method described previously (Blanco et al., 1988). The biochemical assay for the [³H] choline accumulation and [³H] acetylcholine synthesis and release were performed according to the method described previously (Mallick et al., 1994).

Individually collected fractions were assayed for radioactivity and then analyzed by high performance liquid chromatography for [^3H]-acetylcholine as described previously (Potter et al., 1983).

Amounts of radioactivity of [^3H]-acetylcholine release were expressed as femtomoles per milligram (fmol/mg). Drugs used were methyl-3[H]-choline chloride (80m Ci/m mol), acetyl-1-14c-acetyl choline iodide (4.7 m Ci/m mol). Physostigmine salicylate, acetylcholine chloride and choline chloride.

The vibration perception threshold test was performed with a bio thesiometer (Padma et al., 1986). Penile biothesiometer is a simple and efficacious screening test for sensory deficit erectile dysfunction. For the experimental purposes a portable hand-held electromagnetic vibration device was used in which frequency of vibration is fixed but the amplitude of vibration is variable. The study was performed by asking the patients to inform the first sensation of vibration as the amplitude of vibration is slowly increased. A similar amplitude was also observed as the patient was asked to inform the instance. The sensation of vibration disappears as the amplitude of vibration is slowly decreased. The vibration perception threshold values were obtained from the pulp of the index fingers bilaterally, the shaft of the penis bilaterally and the glans penis. The values for the right and left sides of the mid shaft and the right and left index fingers were averaged. Statistical evaluation of the data was made by unpaired t-test for the comparison of the two groups. Significance was considered at $p < 0.05$.

RESULTS

A summary of the results obtained from the biochemical assay of the [^3H] choline accumulation and [^3H] acetyl choline synthesis and release in tissue from IDDM and NIDDM men with and without neuropathy and from non-diabetic controls is presented in Table 2. This study revealed a significant decrease ($p < 0.05$) in the study parameters in corporeal tissue from both IDDM and NIDDM diabetic men with neuropathy in comparison with that from non diabetic controls. However tissue from both types of diabetic patients without neuropathy did not differ significantly in any of the above mentioned parameters from that of non-diabetic controls. No significant difference was apparently observed between IDDM and NIDDM diabetics (with and without neuropathy) in either [^3H] choline accumulation or [^3H] acetyl choline synthesis. In addition, both IDDM and NIDDM diabetic men with neuropathy showed a significant decrease in [^3H] acetyl choline release during transmural electrical field stimulation in comparison with both types of non-neuropathic diabetics and non-diabetic controls. Penile vibration perception threshold values in the mid penile shaft or glans penis did not correlate significantly with either [^3H] choline accumulation or

the [^3H] acetylcholine synthesis in corporeal tissue. Both IDDM and NIDDM diabetic men with neuropathy showed a significant increase ($P < 0.05$) in the value of penile vibration sensitivity in the glans penis compared with both types of non-neuropathic diabetics and non diabetic controls (Table 3).

DISCUSSION

In the present study the corporeal tissue has been used to assess the functional status of penile cholinergic nerves in both IDDM and NIDDM diabetic impotent men (with and without an objective evidence of neuropathy) compared with non diabetic impotent control subjects.

These results indicate a significant decrease ($P < 0.05$) in the values of [^3H] choline accumulation and [^3H] acetylcholine synthesis and release in corporeal tissue from both IDDM and NIDDM diabetic men with neuropathy as compared to non-neuropathic (both IDDM and NIDDM) diabetics and non diabetic controls. However no significant differences between the values of study parameters were observed when these values were compared among IDDM and NIDDM diabetic men with or without neuropathy (Table 3). As the impaired ability of penile cholinergic nerves to synthesize and release acetylcholine was found to have the highest significant difference between neuropathic and non neuropathic and non-diabetic control subjects, the present findings indicate that a primary defective neuropathic mechanism in the cholinergic nerves of penis may play an etiological role in the pathogenesis of erectile impotence in these patients. Although it is not feasible to obtain corporeal tissue from potent diabetic men as a control group, however since penile smooth muscle is necessary for erection (Lue and Tanagho, 1987) and a cholinergic neuro-effector system facilitates this smooth muscle relaxation (Saenz et al., 1988) we assume that the dysfunction of penile cholinergic nerves is contributory to the development of impotence in these diabetic neuropathic men.

Studies have demonstrated morphological alterations in penile unmyelinated nerves (Faerman et al., 1974, Fani et al., 1983, Johnson et al., 1991), and a decrease in the number of the nerves (Crowe et al., 1983, Lincoln et al., 1987) and tissue content of neurotransmitter (Crowe et al., 1983, Gu et al., 1984) in the corpus cavernosum of the diabetic patients and experimental diabetic animals (Mallick et al., 1994). Furthermore diabetic impotent men have a functional impairment of the neurogenic dilator mechanism, which relaxes penile smooth muscle (Saenz et al., 1988). A decrease in the number of acetyl cholinesterase-containing nerves have also been reported in the literature (Lincoln et al., 1987), suggesting a probable loss of some of the cholinergic nerves fibers. The present data is the result of a functional assay in which cholinergic nerves in corporeal tissue synthesize [^3H] acetylcholine from [^3H]

choline. These results however do not provide any evidence regarding decrease in the number of cholinergic nerve terminals or a functional deficit of each of these nerves in these patients.

A biochemical basis for the synthesis of Acetylcholine has been described previously in the nervous tissue (Wike et al., 1993). Acetylcholine is synthesized as a reaction between choline and choline acetyl transfers which is catalysed by acetyl Co A. A direct correlation between the acetylcholine and vasoactive intestinal polypeptide (VIP) in the nerve terminals in the corpus cavernosum tissue has been reported previously (Gu et al., 1984, Lincoln et al., 1987, Yashiastu et al., 1992). It is also evident from the literature that cholinergic nerves via a nor- adrenergic nor-cholinergic neuro transmitter potentiate smooth muscle relaxation (Saenz et al., 1988, Robert et al., 1994), suggesting a complementary role of these neuro-transmitters in the release of penile smooth muscle. An impairment in ability of penile smooth muscle to relax, may therefore be due to the dysfunction of cholinergic nerves along with a decrease in the tissue contents and number of vasoactive intestinal polypeptide immuno-reactive nerves. Similar suggestions have also been reported previously (Crowe et al., 1983, Gu et al., 1984, Lincoln et al., 1987).

A negative correlation between diabetic neuropathic condition and the ability of the cholinergic nerves to the synthesis of acetylcholine in the present study therefore suggest that diabetic patients (both IDDM and NIDDM) with an objective evidence of neuropathy have an abnormal neurogenic smooth muscle relaxation compared to non-neuropathy have an abnormal neurogenic smooth muscle relaxation compared to non-neuropathic IDDM and NIDDM diabetics and non-diabetics impotent men. These results further show an abnormal vibration perception threshold, only in the diabetics with neuropathy. However no significant difference was found when these values were compared between IDDM and NIDDM,diabetic men with or without neuropathy respectively (Table 3). A direct correlation between the functional impairment of penile autonomic cholinergic nerves and an abnormal vibration perception threshold exclusively in both IDDM and NIDDM diabetic patients with neuropathy thus suggests that both sensory and autonomic nerves are dysfunctional in these patients and appears to be a strong predictor of penile neuropathic conditions.

The results of the present study thus provide evidence that an underlying neuropathic mechanism in the cholinergic nerves of penis in both IDDM and NIDDM neuropathic diabetics may result in impaired penile smooth muscle relaxation leading to impotence irrespective of their type of diabetes.

Table 1
Clinical manifestation of autonomic neuropathy

Subjects	Age (Mean ± S.E.)	Impotence (Years) (Mean ± s.e.)	Epigastric Fullness	Postural Hypo- tension	Heart Rate Beat to Beat variation/ Min. (Mean ± s.e.)	Nocturnal Diarrhea
IDDM Without Neuropathy (N = 15)	55 ± 4.4	2.4 ± 0.6	0/15	0/15	19.5 ± 7.9	1/15
IDDM with Neuropathy (N = 15)	56 ± 4.5	2.4 ± 0.7	13/15	15/15	7.9 ± 5.5	13/15
NIDDM without Neuropathy (N = 15)	58 ± 4.0	2.5 ± 0.5	0/15	0/15	19.8 ± 7.5	0/15
NIDDM with Neuropathy (N = 15)	58 ± 2.5	2.6 ± 0.6	14/15	14/15	7.8 ± 5.4	14/15
Non- Diabetic Control (N = 15)	58 ± 4.5	2.5 ± 0.5	0/15	0/15	17.2 ± 7.5	0/15

IDDM = Insulin dependent diabetes mellitus

NIDDM = Non-insulin dependent diabetes mellitus

N = Total Number of subjects studied

Table 2

Tritiated choline accumulation and tritiated acetylcholine synthesis and release in the corporeal tissue of IDDM and NIDDM diabetic men (with and without neuropathy) and in age-matched non-diabetic controls. All values are given as $\pm$ standard error

Subjects	[³ H] choline accumulation (fmol/mg)	[³ H] acetylcholine synthesis (fmol/mg)	[³ H] acetylcholine release (fmol/mg/6/mins)
IDDM without neuropathy (n = 15)	946 $\pm$ 64	245 $\pm$ 35	18.2 $\pm$ 6.8
IDDM with neuropathy (n = 15)	750 $\pm$ 55*	70 $\pm$ 11*	3.0 $\pm$ 0.95*
NIDDM without neuropathy (n = 15)	940 $\pm$ 65	240 $\pm$ 36	17 $\pm$ 6.5
NIDDM with neuropathy (n = 15)	755 $\pm$ 59*	71 $\pm$ 12*	3.5 $\pm$ 0.95*
Non-diabetic controls (n = 15)	945 $\pm$ 68	245 $\pm$ 35	18 $\pm$ 7

n = Total number of subjects studied

IDDM = Insulin dependent diabetes mellitus

NIDDM = Non-insulin dependent diabetes mellitus

IDDM and NIDDM (with and without neuropathy) values are compared with age-matched non-diabetic controls by t-test.

* = P < 0.05

Table 3
Mean vibration perception threshold values of mid penile shaft, glans penis and the pulp of index finger in IDDM and NIDM diabetic men (with and without neuropathy) and in age-matched non-diabetic controls.

Subjects	Mean vibration perception threshold (Mean $\pm$ S.E.)		
	Index Finger	Mid shaft of the penis	Glans penis
IDDM without neuropathy (n = 15)	4.2 $\pm$ 0.4	6.8 $\pm$ 0.8	7.9 $\pm$ 1.4
IDDM with neuropathy (n = 15)	8.8 $\pm$ 1.6*	13.2 $\pm$ 3.5*	15.8 $\pm$ 3.2*
NIDDM without neuropathy (n = 15)	4.5 $\pm$ 0.2	7.1 $\pm$ 0.8	8.5 $\pm$ 1.2
NIDDM with neuropathy (n = 14)	8.5 $\pm$ 1.5*	13.5 $\pm$ 3.2*	16.5 $\pm$ 3.5*
Non-diabetic controls (n = 15)	4.2 $\pm$ 0.2	7.2 $\pm$ 0.8	8.2 $\pm$ 1.4

n = Total number of subjects studied

IDDM = Insulin dependent diabetes mellitus

NIDDM = Non-insulin dependent diabetes mellitus

IDDM and NIDDM (with and without neuropathy) values are compared with age-matched non-diabetic controls by t-test.

* = P < 0.05

REFERENCES

- Ali S.T., Shaikh R.N., Siddiqui N.A., Siddiqui P.Q.R. (1993). *Arch. Andrology*. 30(3): 137.
- Anderson P.O., Bloom S.R., Mellander S. (1984). *J. Physiol.* (London), 351: 209.
- Azadzoï K.M., Saenz de Tajada I. (1992). *J. Urol.* 148: 1587.
- Bart L.H.B., Eric J.H.M., Wim H.D., Servaas L.H.N., Frans M.J.D. (1994). *J. Urol.* 151: 884.
- Blanco R., Saenz deTejada I., Goldstein I., Krane R.J., Wotiz H.H., Cohen R.A. (1988). *Am. J. Physiol.* 254: 468.
- Christ G.J., Schwartz C.B., Stone B.A., Parker M. (1992). *Am. J. Physiol.* 263: 15.
- Crowe R., Lincoln J., Blacklay P.F., Pryor J.P., Lumley J.S.P., Burnstock G. (1983). *Diabetes*. 32: 1075.
- deGroat W.C., Booth A.M. (1980). *Ann. Intern. Med.* 92: 329.
- Diederichs W., Steif C.G., Lue T.F., Tanagho E.A. (1990). *J. Urol.* 143: 1264.
- Ellenberg M. (1980). *Ann. Intern. Med.* 92: 331.
- Faerman I., Glocher L., Fox D., Jadzinsky M.N., *Diabetes*. 23: 971.
- Fani K., Lundin A.P., Beyer M.M., Jimenez F.A., Friedman E.A. (1983). *Diabetologia*. 25: 424.
- Gu J., Polak J.M., Probert L., Islam K.N., Marangos D.J., Bloom S.R. (1983). *J. Urol.* 130: 386.
- Gu J., Polak J.M., Lazarides M., Morgan R., Pryor J.P., Marangos P.J., Blank M.A., Bloom S.R. (1984). *Lancet*. 2: 315.
- Hedlund H., Andersson K.E. (1985). *Acta. Physiol. Scand.* 124: 413.
- Johnson R.D., Halata Z. (1991). *J. Comp. Neurol.* 312: 229.
- Knispel H.H., Goessl C., Beckman R. (1992). *Urol. Res.* 20(4): 253.
- Korenman S.G., Viosca S.P. (1993). *J. Am. Geriatr. Soc.* 41: 363.
- Lehman T.P., Jacobs J.A. (1982). *J. Urol.* 129: 291.
- Lincoln J., Crowe R., Blacklay P.F., Pryor J.P., Lumley J.S.P., Burnstock G. (1987). *J. Urol.* 13: 1053.
- Luangkhot R., Rutchik S., Agarwal V., Puglia K. (1992). *J. Urol.* 148: 467.
- Lue T.F., Tanagho E.A. (1987). *J. Urol.* 137: 829.
- Mallick H.N., Manchandra S.K., Komar V.M. (1994). *J. Urol.* 151: 759.
- Padma-Nathan H., Goldstein I., Krane R.J. (1986). *Sem. Urol.* 4: 225.
- Potter P.E., Mack J.L., Neff N.H. (1983). *J. Neurochem.* 41: 188.
- Robert D., Hoeldtke., Guenther B. (1994). *Ann. Intern. Med.* 120: 512.
- Sae C.K., Moon M. (1992). *J. Urol.* 147: 1530.
- Saenz de Tajada I., Blanco R., Goldstein I., Axadzoï K., delas Morenas A., Krane R.J., Cohen R.A. (1988). *Am. J. Physiol.* 254: 459.
- Wang C.J., Shensy W.U., Huang C.H., Chiang C.P. (1993a). *Urol. Int.* 50: 209.
- Wang R., Higuera T.R., Sikka S.C., Minkes R.K. (1993). *Urol. Res.* 21: 75.

- Weiss H.D. (1992). *Ann. Intern. Med.* 76: 793.
- Wike E.A., Rilay D.A., Lelkes P.I., Hillard C.J. (1993). *Diabetes.* 42: 868.
- Willis E., Ottesen B., Wagner G., Sundler F., Fahrenkrug J. (1981). *Diabetologia.* 20: 281.
- Yashiastu T., Sherif R., Francois B., Christian G., Tom F.L., Emil A.T. (1992). *J. Urol.* 148: 446.